Lab Sheet 03

K-Means Clustering in R

Data Mining — Practical Lab

This lab sheets contains (Introduction to R and K-Means clustering In R). You should be comfortable with vectors, data frames, and basic plotting before starting. In this lab you will use the k-means clustering algorithm to explore two classic datasets: iris and wine.

Learning Outcomes

- Explain the k-means algorithm and its main steps.
- Run k-means clustering in R using the kmeans() function.
- Use NbClust and the elbow method to choose an appropriate number of clusters k.
- Pre-process (scale) data before clustering, and explain why this matters.
- Evaluate a clustering solution against known class labels using a confusion matrix and the Adjusted Rand Index (ARI).
- Visualise clustering results.

Preparation

- Open RStudio and set your working directory to your Lab Sheet 02 folder.
- Make sure the following packages are installed: datasets, NbClust, fpc, MASS, flexclust.
- Download / confirm you have the file winedata.csv in your working directory.

```
install.packages(c("NbClust","fpc","MASS","flexclust"))
```

Part A — K-Means on the Iris Dataset

Question 1. Loading and inspecting the data

(a) Load the built-in datasets library and display the first six rows of the iris dataset.

(b) List the column (variable) names of iris. Which column is the class label, and how many classes does it have?

(c) Create two new variables: x containing only the four numeric feature columns, and y containing the Species column.

Question 2. Running your first k-means model

(a) Fit a k-means model to x using 3 clusters. Store the result in a variable called kc.

(b) Print kc and identify: (i) the size of each cluster, (ii) the cluster means, (iii) the between_SS / total_SS percentage.

(c) Produce a confusion-style table comparing y (true species) with kc$cluster. Which species is perfectly separated from the others, and which two species are most often confused?

Question 3. Visualising the clusters

(a) Produce a scatter plot of Sepal.Length against Sepal.Width, colouring points by their assigned cluster (kc$cluster).

(b) Overlay the cluster centres on the same plot using points() with a distinct plotting symbol.

(c) Repeat the plot using Petal.Length and Petal.Width instead. Which pair of variables gives visually cleaner clusters? Explain briefly why this makes biological sense.

Question 4. Reproducibility and initial centroids

(a) Explain, in your own words, why running kmeans() twice on the same data can give two different results.

(b) Set a random seed of your choice with set.seed(), then re-run k-means on iris[,3:4] with 3 clusters and nstart = 20. What does the nstart argument do, and why is it good practice to use it?

(c) Compare the resulting clustering to the true Species labels using table(). How many observations are misclassified?

Question 5. Choosing the number of clusters

(a) Using the elbow method, compute the total within-cluster sum of squares (tot.withinss) for k = 2 to k = 10 on the iris feature data, and plot k against the within-cluster sum of squares.

(b) From your plot, identify the k value at the 'elbow' (bend) of the curve. Does this agree with the known number of iris species?

(c) Load the NbClust package and run NbClust() on the iris feature data (with the Species column removed) for min.nc = 2, max.nc = 10, method = "kmeans", index = "all". According to the majority rule, what is the best number of clusters?

(d) Briefly explain, in 2–3 sentences, why a single index (like the elbow plot) might disagree with the majority-vote result from NbClust, and why it can be useful to consult multiple indices.

Part B — K-Means on the Wine Dataset

Question 6. Loading and exploring winedata.csv

(a) Read winedata.csv into a data frame called wine using read.csv().

(b) Use str() to inspect the structure of wine. How many observations and how many variables are there? Which variable represents the wine's true class (cultivar)?

(c) Using summary() or otherwise, comment on whether the 13 chemical variables are all measured on similar scales.

Question 7. Scaling the data

(a) Create a scaled version of the chemical variables (excluding the Type column) called data.train, using the scale() function.

(b) Explain why scaling is necessary before applying k-means to this dataset, referring to how k-means computes distances.

(c) Confirm that each scaled variable now has a mean of (approximately) 0 using summary(data.train) or colMeans().

Question 8. Choosing k and fitting the model

(a) Run NbClust on data.train (min.nc = 2, max.nc = 15, method = "kmeans") and produce a bar chart of the number of criteria proposing each value of k.

(b) Separately, write a for loop that computes the within-cluster sum of squares for k = 1 to 15 and plot the resulting scree plot. Does the elbow agree with NbClust's recommendation?

(c) Set a seed and fit a final k-means model, fit.km, to data.train using the number of clusters you decided on in parts (a)-(b).

Question 9. Interpreting and evaluating the clusters

(a) Print fit.km$centers. Choose one cluster and describe, in a sentence or two, what kind of wine it represents (e.g. relatively high/low in particular chemicals).

(b) Build a confusion matrix comparing wine$Type to fit.km$cluster using table(). How many wines are misclassified overall?

(c) Load the flexclust package and compute the Adjusted Rand Index (ARI) for your confusion matrix using randIndex(). What does an ARI close to 1 indicate, and what would an ARI of 0 mean?

Question 10. Visualisation

(a) Using the fpc package, produce a discriminant projection plot of the wine clusters with plotcluster().

(b) In 2–3 sentences, comment on how well-separated the three clusters appear to be in this projection.

Part C — Short Answer / Theory

Question 11. Conceptual questions

(a) List, in order, the four main steps of the k-means algorithm.

(b) K-means is described as a 'partitioning' method. Name one clustering method that is NOT a partitioning method, and briefly state one key difference between it and k-means.

(c) Give one advantage and one limitation of k-means clustering compared to hierarchical clustering.

(d) Why is k-means considered an unsupervised learning technique, even though the datasets you used in this lab (iris and wine) actually have known class labels?

Please attempt all questions in Parts A, B and C. Show your R code and output (plots and console output) for every sub-question in your submission.